

Guide to Water Free Lithium Bis(oxalate) Borate (LiBOB)

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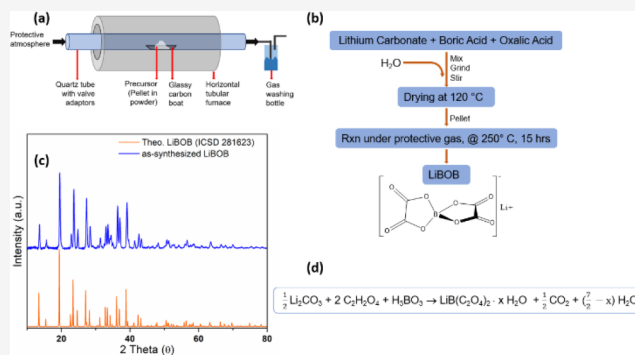
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ABSTRACT: Lithium bis(oxalate) borate, $\text{LiB}(\text{C}_2\text{O}_4)_2$ (LiBOB), is one of the most important electrolyte additives for Li-ion batteries (LIBs) due to its numerous advantages such as thermal stability, good solubility in organic solvents, high conductivity, and low cost as well as providing safer operations with superior electrochemical performance compared to conventional electrolyte combinations. However, the use of LiBOB is limited due to slight instability issues under ambient conditions that might require extra purification steps and result in poorer performances in real systems. Here, we address some of these issues and report a high purity water free LiBOB synthesized with fewer processing steps, employing lithium carbonate, oxalic acid, and boric acid as low-cost starting materials, and via ceramic processing methods under protective atmosphere. The physical and chemical characterizations of both anhydrous and monohydrate phases are performed with X-ray powder diffraction (XRPD), Fourier-transform infrared spectroscopy (FTIR), Raman spectroscopy, and scanning electron microscopy (SEM) analyses to determine the degree of the purity and the formation of impurities, such as $\text{LiBOB} \cdot \text{H}_2\text{O}$, HBO_2 , and $\text{Li}_2\text{C}_2\text{O}_4$, as a result of the aging investigations where the as-synthesized salt was exposed to ambient conditions for different durations. Differential thermal analysis (DTA) is applied to determine the optimum synthesis conditions for anhydrous LiBOB and to analyze the water loss and the decomposition of $\text{LiBOB} \cdot \text{H}_2\text{O}$. Aging experiments with the water free LiBOB are carried out to evaluate the effect of humidity on the phase changes and resulting impurities under various conditions. The detrimental effect of even slightest humidity conditions is shown, and protective measures during and after the synthesis of LiBOB are discussed. Anhydrous LiBOB could be widely used as an electrolyte additive to improve the overall electrochemical performances for LIBs through development of a protective solid electrolyte interface (SEI) on the surface of high voltage cathodes and by bringing about superior electrochemical properties with increased cycling stability, rate capability, and Coulombic efficiency, if synthesized, purified, and handled properly before use in real electrochemical systems.



INTRODUCTION

Electrolyte salts and additives play a crucial role in increasing the cycle life, capacity, and safety of batteries especially for high voltage applications.^{1–3} Lithium bis(oxalate) borate, $\text{LiB}(\text{C}_2\text{O}_4)_2$ (LiBOB), is one of the most widely used electrolyte additives in Li-ion batteries and has the potential to be an alternative for the commercially used lithium hexafluorophosphate (LiPF_6) as an electrolyte salt when dissolved in suitable solvents. LiBOB provides electrochemical stability at high applied potentials and high thermal stability under various electrochemical conditions.^{4–6} LiBOB also promotes the formation of a protective solid–electrolyte interface (SEI) layer, thus increasing capacity and cycle life when used as an electrolyte additive.^{7–14}

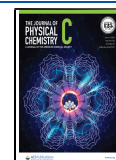
LiBOB can address the need for stable electrolyte compositions for high voltage cathode materials and applications. In a mixed solvent system (γ -butyrolactone and dimethyl carbonate), it was shown that LiBOB provides oxidative stability up to 5.3 V in Li-ion batteries,^{3,15} whereas LiPF_6 can be only employed up to 4.5 V. The thermal stability

of bulk LiPF_6 is 80 °C, and it is safe to use up to 50 °C in LIBs. At higher temperatures, LiPF_6 is susceptible to decompose into LiF and PF_5 , resulting in low electrochemical performances and safety issues.^{8,16} On the other hand, bulk LiBOB decomposes at 300 °C, and it is stable up to 70 °C in LIBs.¹⁷ There are several other reports where LiBOB enables stable operation at voltages higher than 4.5 V.^{18–20} Besides this, LiBOB provides a safer operation for Li-ion batteries not only due to its high thermal stability but also due to its role in protective SEI formation. SEI layer formation does not only play an active role only in the first cycle in the battery, yet SEI keeps growing and changing with processes such as dissolution and precipitation.^{21,22} Thus, understanding the SEI chemistry

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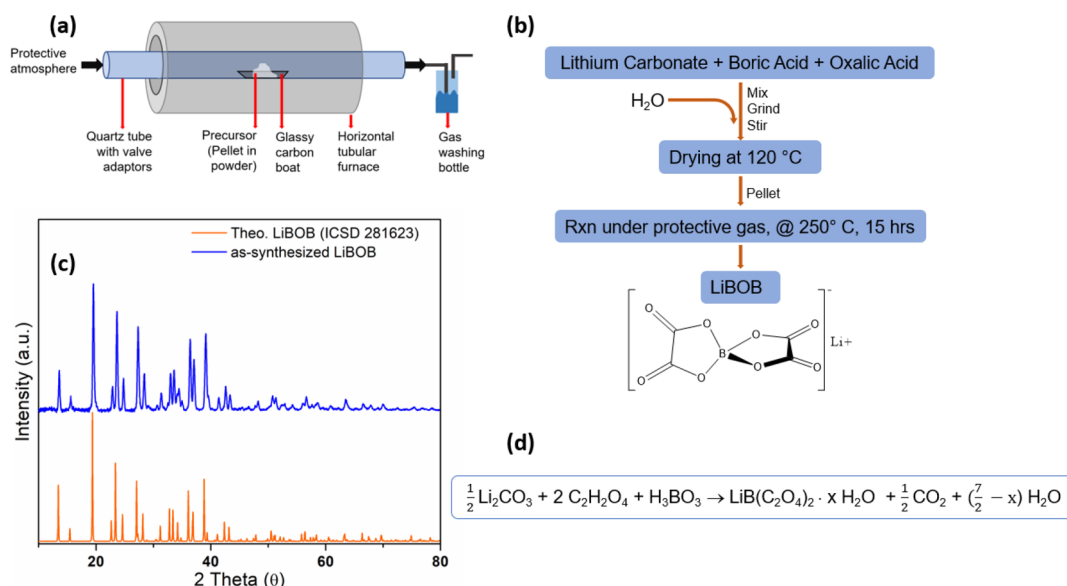


Figure 1. (a) Schematic representation of the experimental setup for the high purity anhydrous LiBOB synthesis. Ar, N₂, and dry air are employed as protective gases (please see Figure S1 for the N₂ and dry air synthesis results). (b) Processing steps used in the synthesis of high purity anhydrous LiBOB. (c) XRPD patterns of theoretical LiBOB (ICSD 281623)³⁹ (orange) and as-synthesized LiBOB (blue). Reflections belonging only to LiBOB are observed in the XRPD pattern for the synthesized sample. (d) Overall chemical equation for the LiBOB synthesis reaction.

and stabilizing the SEI are important challenges in the development of long-life and safe Li-ion batteries, as well.

Other common Li-ion battery salts such as LiPF₆, LiAsF₆, and LiClO₄ were shown to decompose to form corrosive and reactive chemicals.²³ The high oxidation state of Cl(VII) in LiClO₄ causes reactions with a range of organic solvents,²⁴ As(III) and As(0) that form during redox reactions involving LiAsF₆ are toxic,²⁵ and LiTf-based electrolytes are highly corrosive.²³ On the other hand, LiBOB has received great interest owing to its superior properties such as good solubility in organic solvents, improved cycling stability with high voltage cathodes, environmentally friendliness, and low cost.^{9,11,14,26,27} Additionally, other potentially corrosive and dangerous byproducts such as HF, PF₅, and LiF do not evolve upon cycling when LiBOB is used.^{28–30} LiBOB was also shown to protect graphite anodes from exfoliation in PC-based electrolytes and to form a SEI on the graphitic anode surface.⁶ Unlike LiPF₆ and other salts such as LiClO₄, LiAsF₆, and LiTf, LiBOB directly forms a protective SEI independent of the carbonate electrolyte solvent employed. Turning to the applications of LiBOB as an additive, Pieczonka et al. observed that the addition of 1 wt % LiBOB in traditional electrolytes increases the cycling stability and Coulombic efficiency for a high voltage spinel/graphite battery system, and the addition of 3 wt % LiBOB eliminated the PF₅, decreasing the Mn dissolution from the Li_{1-x}Ni_{0.42}Fe_{0.08}Mn_{1.5}O₄ (LNFMFO) cathode.¹⁹ In another study, Lee and co-workers employed LiBOB as an electrolyte additive to develop a protective SEI for high potential Li_{1.17}Ni_{0.17}Mn_{1.5}Co_{0.17}O₂ cathodes improving the cycling stability and rate capability via eliminating the electrolyte decomposition.³¹

Although LiBOB is a very promising electrolyte salt and additive for Li-ion batteries, to date, there are only few reports^{32–34} investigating different synthetic routes. Yu et al. employed a solid-state reaction for the first time using oxalic acid, lithium hydroxide, and boric acid as starting materials followed by a thermal treatment in air and a recrystallization process to obtain LiBOB; in addition, they also reported that

LiBOB decomposes into Li₂C₂O₃, B₂O₃, CO₂, and CO at temperatures higher than 300 °C.^{32,35} In another study, Lestariningsih and co-workers utilized lithium carbonate instead of lithium hydroxide to eliminate the LiBOB·H₂O phase. However, they improved the purity of anhydrous LiBOB only up to 59.1% using Li₂CO₃ instead of LiOH which yielded only 17.2% pure anhydrous product.^{34,36} Lian and his group performed a rheological phase reaction using oxalic acid dihydrate, lithium hydroxide monohydrate, and boric acid. They reported that LiBOB synthesized from a solid-state reaction may suffer from impurities requiring additionally recrystallization steps.³³ In addition, to the best of our knowledge, LiBOB·H₂O is present as a byproduct in all reported LiBOB synthesis methods in the literature. Depending on the reactants used in the synthesis, impurities such as lithium oxalate and lithium carbonate were also detected in the final product. Moreover, there is a lack of research focusing on the stability of this important electrolyte salt and additive under humidity during and after the synthesis, and under prolonged air exposure.^{37,38} Therefore, herein, we focus on these issues and report on the importance of new synthetic routes and humidity conditions to obtain a high purity water free LiBOB. We synthesize anhydrous and higher purity LiBOB using low cost starting materials and fewer processing steps compared to most prevalent synthesis methods in the literature. Avoiding moisture and impurities in LiBOB and such electrolyte salts are vital for the performance of LIBs. Thus, we also underline the importance of synthesizing and employing LiBOB under protective atmosphere to eliminate the formation of its hydrous phase and impurities by studying the aging of the synthesized high purity salt under atmospheric conditions.

EXPERIMENTAL SECTION

Synthesis of Anhydrous LiBOB. LiBOB is synthesized using lithium carbonate (Li₂CO₃, ≥99%, Alfa Aesar), oxalic acid (C₂H₂O₄, ≥98%, Alfa Aesar), and boric acid (H₃BO₃,

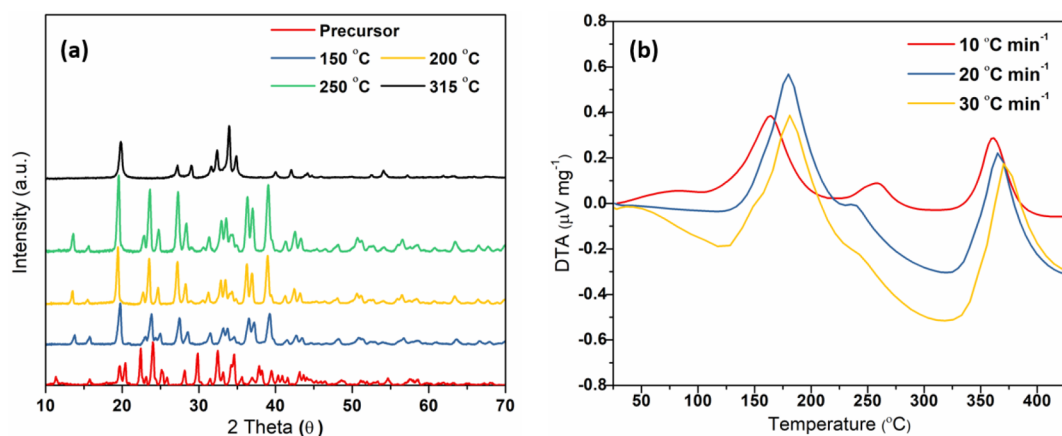


Figure 2. (a) XRPD patterns for the precursor and samples synthesized at 150 (blue), 200 (yellow), 250 (green) and 315 °C (black). (b) DTA data taken at different heating rates showing the reaction temperature for LiBOB synthesis and decomposition temperature of the synthesized LiBOB.

≥99%, Alfa Aesar) as starting materials. Stoichiometric amounts of the compounds (following the equation in Figure 1d) are mixed and ground in a mortar. An excess amount of deionized water is added to the powder mixture, and the mixture is stirred continuously at 80 °C under ambient conditions, in a fume hood exposed to air, until all water is evaporated. A pellet made from this precursor powder is placed in a glassy carbon boat and covered/protected with the same powder. Then, the reaction boat is placed in a quartz tube within a horizontal tubular furnace, and the reaction is carried out under protective atmosphere (Ar, N₂ or dry air) at 250 °C for 10–15 h. The products are transferred into an Ar glovebox (O₂ and H₂O < 1 ppm) without exposure to ambient atmosphere. After the synthesis, the protective powder is discarded, and the anhydrous LiBOB pellet is ground in a mortar in the glovebox. Aging of the anhydrous LiBOB powder is done under ambient conditions (at RT (~25 °C), and 70% average relative humidity). Then, 2–3 g of LiBOB powder is placed in semiclosed autoclave bottles and left in fume hoods with ventilation for 5, 15, and 30 min and 2 days. The main aim of the aging experiments is to determine the effect of humidity on storing, handling, and using LiBOB in air and to stress the importance of synthesis under protective conditions.

Characterization of LiBOB. The phase and purity of the samples are determined by X-ray powder diffraction (XRPD) method using a Bruker D8 Advance X-ray diffractometer, and Cu Kα₁ radiation (operated at 40 mA, 40 kV). The purity of the as synthesized LiBOB and impurities forming upon aging are further characterized by vibrational spectroscopy analyses. Fourier-transform infrared (FTIR) spectroscopy is performed on a Thermo Scientific Nicolet iS10 spectrometer with a single reflection diamond attenuated total reflectance (ATR) module, and Renishaw inVia Raman microscope equipped with a 532 nm laser as the excitation source is used to perform Raman spectroscopy. The morphology of the as synthesized and aged powders is investigated by scanning electron microscopy (SEM) analysis using a Zeiss Ultra Plus field emission GeminiSEM. Differential thermal analysis (DTA) is utilized to monitor the synthesis and any exothermic and endothermic reactions in the selected temperature range using a Netzsch STA 449 thermal analysis system. For the XRPD measurements, as synthesized anhydrous LiBOB samples are prepared in an Ar glovebox, and the sample holders are covered with Kapton tape to prevent exposure to air. No special precaution

is taken for SEM, FTIR, and Raman investigations. Samples are put in a sealed protective container in an Ar glovebox and taken to the measurement stages of the instruments. For the SEM analyses, samples were directly transferred to the vacuum chamber of SEM after being removed from the sealed container. For FTIR and Raman spectroscopies, the anhydrous samples were exposed to air briefly during the measurement after being removed from the sealed sample holders.

RESULTS AND DISCUSSION

LiBOB is synthesized in high purity following the procedure depicted in Figure 1, parts a and b. The phase and purity of as-synthesized LiBOB is investigated via XRPD analysis by comparing the experimental reflection pattern with the calculated reflection pattern of LiBOB (ICSD 281623) (Figure 1c). The XRPD pattern of the as-synthesized LiBOB matches the theoretical pattern, and the reflections belong only to anhydrous LiBOB.³⁹ This finding shows that single phase anhydrous LiBOB could be obtained in a single synthetic step without the need of any extra recrystallization steps to eliminate LiBOB·H₂O. The lack of any other reflections belonging to possible impurity phases shows that they are under ~1 wt %. Most importantly, LiBOB·H₂O, which is the main impurity in the synthesis that could also cause parasitic reactions in LIBs, is not observed. Key points in achieving high purity anhydrous LiBOB with this experimental design were conducting the reaction under inert atmosphere (Ar, N₂ or dry air) and protecting the pellet with the precursor powder following well-known ceramic processing methods (Figure 1, parts a and c and Figure S1).

The reaction of the precursor pellet was monitored via differential thermal analysis (DTA), and the synthesis was also carried out at various temperatures to determine the optimum temperature window for the high purity LiBOB synthesis (Figure 2). From the XRPD analysis and DTA data, 225–300 °C range could be claimed to be the optimum temperature range for this synthesis. The XRPD and DTA findings show that anhydrous LiBOB decomposes into lithium oxalate (Li₂C₂O₄) and other amorphous phases at temperatures higher than ~315 °C, and at temperatures below ~250 °C, either monohydrate phase is observed or the reaction is not complete (Figure 2a and Figure S5). According to the reference intensity ratio (RIR) method,^{40–42} the percentages of anhydrous LiBOB

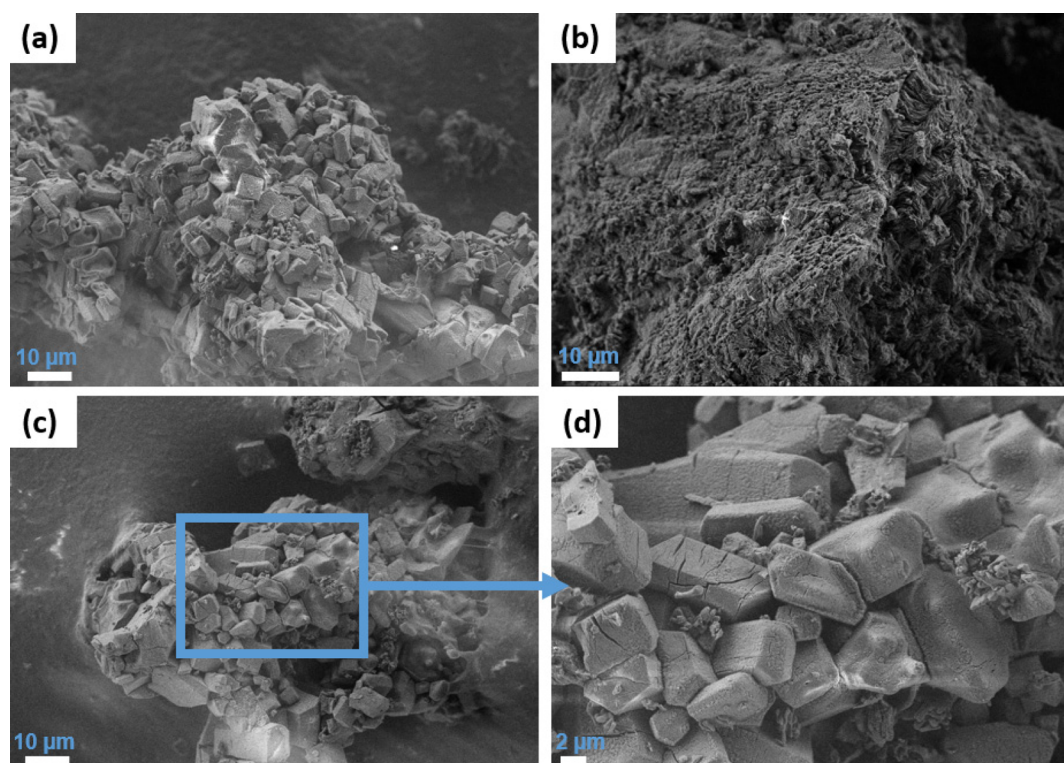


Figure 3. (a, c) SEM micrographs showing the particle size and morphology of anhydrous LiBOB synthesized at 250 °C. (b) Morphology of the products synthesized at 150 °C. (d) Magnified region from part c, displaying LiBOB crystals.

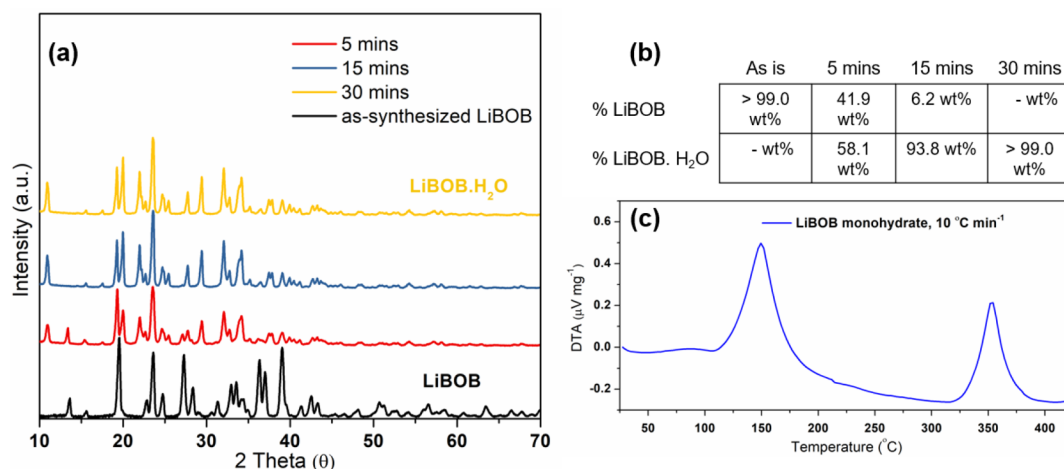


Figure 4. (a) XRPD patterns of the as-synthesized and aged (5, 15, and 30 min) LiBOB samples. (b) Phase composition of LiBOB and LiBOB·H₂O in as-synthesized and aged samples calculated via reference intensity ratio (RIR) method. (c) DTA analysis showing the water loss and decomposition of LiBOB·H₂O.

for as-synthesized samples at 200 and 250 °C were found to be ~95 and ~99.0 wt %, respectively. Humidity and impurities in LiBOB have detrimental effects on the performance of Li-ion batteries,⁴³ and although it was claimed that monohydrate phase of LiBOB is stable,⁴⁴ we show that hydrated LiBOB could decompose into HBO₂ and Li₂C₂O₄ when exposed to air (Figure S2). The presence of such impurities in electrolyte solution could cause capacity fading and safety problems under extensive cycling.

Scanning electron microscopy (SEM) was employed to further understand the particle morphologies and the effect of humidity on the morphology. A homogeneous distribution of particles is observed in SEM images for the samples calcined at

temperatures between 250–300 °C (Figure 3a,c,d). These anhydrous LiBOB particles are few micrometers in size and prism shaped, whereas synthesis at lower temperatures (<150 °C) and the product after prolonged air exposure yield rumbled morphologies without any well-defined structures. (Figure 3b and Figure S4). The detrimental effect of humidity on high purity LiBOB can be easily observed via SEM analysis as well (Figure S3). The morphology of the anhydrous LiBOB particles start changing after ≤5 min-long exposure to air, and curvatures and pores start to form on the surface of prisms (Figure S3).

In order to analyze the effect of prolonged air exposure on LiBOB, and to further investigate the influence of humidity in

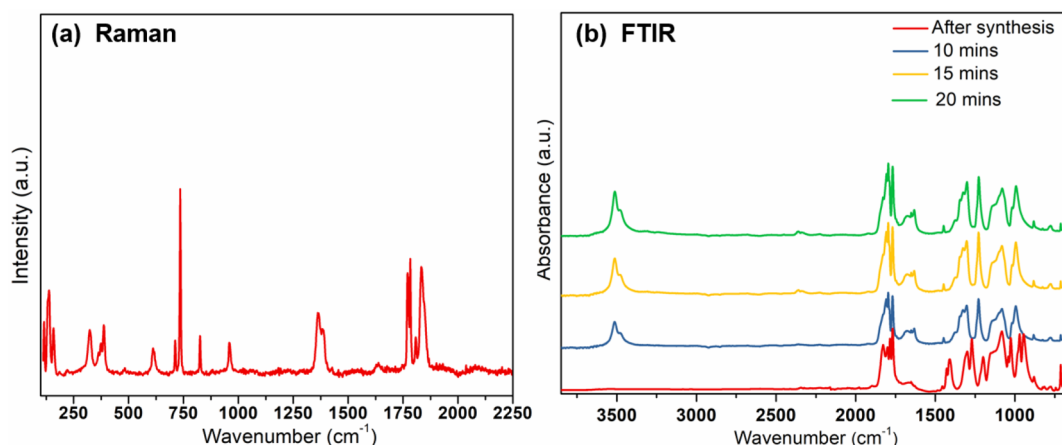


Figure 5. (a) Raman spectra of anhydrous LiBOB. (b) FTIR spectra of as synthesized and aged LiBOB showing the formation of monohydrate phase and secondary phases upon exposure to ambient conditions, i.e., humidity in air.

phase changes, aging experiments were performed. Aging of the anhydrous LiBOB powder was done under ambient conditions (at RT (~ 25 °C), and 70% average relative humidity) for 5, 15, 30 min and 2 days. The effect of humidity is first investigated through XRPD analysis. XRPD powder patterns in Figure 4a show that as-synthesized powder consists of only anhydrous LiBOB phase, and ~ 42 wt % monohydrate phase ($\text{LiBOB} \cdot \text{H}_2\text{O}$) forms after 5 min of aging under given conditions according to the RIR method. After only 15 min of exposure to air under given conditions, almost all of the anhydrous LiBOB turns into $\text{LiBOB} \cdot \text{H}_2\text{O}$, and the complete hydration occurs after 30 min of aging (Figure 4a, Figure S5, and Figure 4b). The presence of water in the structure can be further confirmed by DTA analysis (Figure 4c). The structural water is given from the monohydrate phase at ~ 150 °C, and the phase decomposition into $\text{Li}_2\text{C}_2\text{O}_4$ and other amorphous phases starts at ~ 315 °C. These findings signify the importance of synthesizing and preserving LiBOB under a protective atmosphere. The thermogravimetric analysis (TGA) also corroborates these points and the lack of monohydrate phase if the synthesis and preservation are done under given conditions (Figure S6). Since purity of electrolyte salts and additives play a crucial role in the cycle life and safety of LIBs,⁴⁵ the measures taken here in the synthesis of anhydrous LiBOB can be considered as critical necessary steps.

The vibrational spectroscopy analysis is in good agreement with XRPD and DTA analyses in terms of confirming the high purity LiBOB phase after the synthesis, and it verifies the humidity effect after aging. Both Raman and FTIR spectra of the as-synthesized phase clearly show the high purity of the as-prepared LiBOB after the synthesis,^{10,43,46} and no major impurity phases are observed (Figure 5, parts a and b). In the FTIR spectrum (Figure 5b) of the LiBOB phase, the strong absorption peak at $1765\text{--}1825$ cm⁻¹ could be assigned to the stretching vibrations of the C=O bond, whereas the region of $980\text{--}1370$ cm⁻¹ is attributed to the C–O and B–O stretching modes.^{10,32} To analyze the hydration of the anhydrous phase, the FTIR spectra from the aged LiBOB samples were also collected. The water absorption peak intensity at ca. 3500 cm⁻¹ increases with the aging time, and several absorption peaks in the range of $1500\text{--}800$ cm⁻¹ shift-change due to the structural deterioration of LiBOB.^{43,47} It should be also noted that a very small water peak might be visible in the spectrum of the as-synthesized sample, since the measurement was not done in a

protective atmosphere, so the sample was exposed to air for a minute. As the aging time increases, it is also evident that there is $\text{Li}_2\text{C}_2\text{O}_4$ formation in samples with the observation of small intensity peaks at 1658 and 778 cm⁻¹.^{43,48} These findings are also consistent with XRPD patterns of the aged LiBOB for 2 days and the decomposed LiBOB (Figure S2, Figure 2a–315 °C pattern). As shown in Figure S2, in addition to the diffraction pattern of the $\text{LiBOB} \cdot \text{H}_2\text{O}$, reflections at 20.08° and 27.88° could be attributed to the formation of HBO_2 , and reflections at 22.06° and 29.48° indicate the presence of $\text{Li}_2\text{C}_2\text{O}_4$, as similarly observed by other investigators.^{33,49}

To sum up, the detrimental effect of humidity during and after the synthesis of LiBOB is evident with the monohydrate phase formation that could further decompose mainly into $\text{Li}_2\text{C}_2\text{O}_4$ and amorphous other phases. These impurities could of course be eliminated through repeated recrystallization steps with polar aprotic organic solvents, but this would bring about additional complications, such as cost increases and environmental burden with the use of organic solvents. Therefore, a high purity synthesis of anhydrous LiBOB with the least possible steps is very crucial, and this would decrease the number of recrystallization steps in case they are needed. The employment and widespread application of LiBOB as an electrolyte additive in LIBs could be considered as limited in the current state-of-art. Yet, we think that this limitation is also present because of the unexpected or low quality results obtained due to the occurrence of impurities and hydrated phases.^{43–45,50} This occurrence and impurities may not even be realized by the unsuspecting, as they are very easy to form as we have shown here. Nevertheless, with the proper use, LiBOB could be expected to bring superior electrochemical performances when it is used as an electrolyte additive for LIBs including improved cycling stability, rate capability, energy density, and Coulombic efficiency by preventing the electrolyte decomposition and the formation of dangerous byproducts.^{31,33,51–53} Additionally, a protective LiBOB-derived SEI will develop on the electrode surface, enhancing the interfacial stability for high temperature operations.^{14,46}

CONCLUSIONS

LiBOB was synthesized under protective atmosphere using lithium carbonate, oxalic acid, and boric acid as low-cost starting materials by applying ceramic processing methods to achieve a high purity product. Aging investigations were also

carried out to further evaluate the effect of humidity and the formation of the monohydrate phase ($\text{LiBOB} \cdot \text{H}_2\text{O}$) and impurities such as HBO_2 and $\text{Li}_2\text{C}_2\text{O}_4$ that are undesirable in applications for high performance LIBs. According to the DTA results, the synthesis temperatures above 315°C cause decomposition of anhydrous LiBOB into impurities; and monohydrate phase that forms at temperatures below 200°C may result in capacity fading and poor cycling performance in LIBs. As shown from the XRPD data and FTIR and Raman spectra, the resulting salt is anhydrous and high purity over $\sim 99\text{ wt } \%$; thus, a higher purity is obtained with fewer and practical processing steps compared to some commercial products and LiBOB reported in the literature. XRPD analysis showed that $\sim 42\text{ wt } \%$ $\text{LiBOB} \cdot \text{H}_2\text{O}$ formed only after 5 min of aging at RT ($\sim 25^\circ\text{C}$)-70% relative humidity, and only after 30 min of exposure to air under the given conditions, the complete hydration occurred. From the SEM micrographs, consistent with the findings from XRPD and FTIR studies, the structural deterioration of LiBOB can also be tracked. The longer exposures in the time scale of days further decomposed $\text{LiBOB} \cdot \text{H}_2\text{O}$ into other impurity phases, mainly $\text{Li}_2\text{C}_2\text{O}_4$. Keeping the product purity high and avoiding exposure to humidity in all processing steps are crucial for stable performance of batteries using LiBOB either as an electrolyte salt or additive and toward widespread applications, especially high voltage LIBs.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.1c01437>.

XRPD patterns of high purity anhydrous LiBOB synthesized using different protective gases, i.e., under dry air and N_2 atmosphere; XRPD pattern showing crystalline water, HBO_2 , and $\text{Li}_2\text{C}_2\text{O}_4$ formation after aging anhydrous LiBOB for 2 days under ambient conditions; SEM micrographs for anhydrous LiBOB aging under ambient conditions showing changes in the prismatic morphology and formation of pores; SEM Micrographs showing the morphology of the products synthesized at $150\text{--}200^\circ\text{C}$; XRPD patterns of theoretical anhydrous LiBOB, as-synthesized anhydrous LiBOB, theoretical $\text{LiBOB} \cdot \text{H}_2\text{O}$, and as-obtained $\text{LiBOB} \cdot \text{H}_2\text{O}$ (30 min air exposed sample); and comparison of TGA curves of the as-synthesized anhydrous LiBOB and $\text{LiBOB} \cdot \text{H}_2\text{O}$ (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Author Contributions

*C.Z. and Y.S. should be considered cofirst authors.

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

DTA, differential thermal analysis; FTIR, Fourier-transform infrared; LiBOB, lithium bis(oxalate) borate; LIBs, lithium-ion batteries; RIR, reference intensity ratio; SEI, solid–electrolyte interface; XRPD, X-ray powder diffraction; SEM, scanning electron microscopy; TGA, thermogravimetric analysis.

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